Law of Equipartition of Energy

Law of Equipartition of Energy states that –

Total energy of a molecule is equally partitioned among its various degrees of freedom and each degree of freedom contributes ½ *k*T amount of energy.

$\epsilon = \frac{3}{2}kT$ (ϵ = Trans. K.E of molecule)

Now, $\epsilon = \epsilon_x + \epsilon_y + \epsilon_z$; $\epsilon_x, \epsilon_y, \epsilon_z$ are components of trans. KE in x-, y- and z-directions, respectively.

$\epsilon_x = \frac{1}{2} m \bar{c}_x^2$

$\epsilon_y = \frac{1}{2} m \bar{c}_y^2$

$\epsilon_z = \frac{1}{2} m \bar{c}_z^2$

But, $\epsilon_x = \epsilon_y = \epsilon_z$

Since, $\epsilon = \frac{3}{2}kT$ Thus, $\epsilon_x = \epsilon_y = \epsilon_z = \frac{1}{2}kT$

Thus, total translational K.E. is equally shared among its various components and each component contributes ½ *k*T amount of energy.

In case of rotation; Rotational KE is expressed as $\frac{1}{2} I \omega^2$

$I \rightarrow$ Moment of inertia

ω = Angular velocity

In case of vibration; $KE = \frac{1}{2}\mu(r - r_{eq})^2$

μ : Reduced mass ; r_{eq} = Equilibrium bond distance.

r : Interatomic distance at any instance.

$PE = \frac{1}{2} k \omega^2$; k = Force const.

ω = Vib. frequency.

vibrational motion along the molecular axis.

$\left[\mu = \frac{m_A m_B}{m_A + m_B} = \text{reduced mass} \right]$

Law of Equipartition of Energy: C_P and C_V Calculation

Degrees of Freedom: **The minimum number of independent coordinates needed to specify the position and configuration of a thermo-dynamic system in space is called degrees of freedom of the system.**

Examples:

1. **A free particle moving along X-axis needs only one coordinate to specify it completely.**
   So, F = 1

2. **A particle moving over a plane has F = 2**

3. **A particle moving in space has F = 3**

Suppose, we have n number of gas molecules in the container, then F = 3n

But, if the system has q number of constraints (restriction in motion), the F decreases to F = 3n-q

In general:

Total degrees of freedom ; $F = 3n$

Trans. " " " ; $F = 3$

Rot. " " " ; $F = 2$ (linear)
$= 3$ (non-linear)

Vib. " " " ; $F = (3n-5)$ (linear)
$= (3n-6)$ (non-linear)

vibrational motion contribution :

$2 \times \frac{1}{2} kT = kT$,

since it has both kinetic and potential energy modes.

Law of Equipartition of Energy: C_P and C_V Calculation

(A) Monoatomic gases :- Total degree of freedom $= 3n$; n = no. of atoms in a molecule

Total degree of freedom $= 3 \times 1 = 3$

Trans. degree of freedom $= 3$

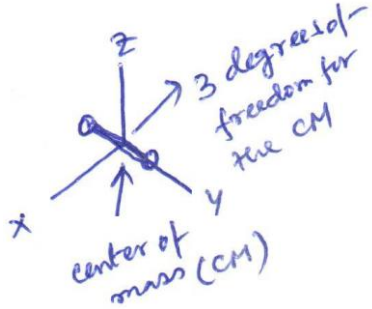
Total energy ; $\epsilon = 3 \times \frac{1}{2} kT$

Total molar energy, $E = 3 \times \frac{1}{2} RT$

$$C_V = \left(\frac{\partial E}{\partial T}\right)_V = \frac{3}{2} R \simeq 3 \text{ cal mol}^{-1} K^{-1}$$

$$C_P = C_V + R \simeq 5 \text{ cal mol}^{-1} K^{-1}$$

(B) Diatomic gas :- Total degrees of freedom, $F = 3 \times 2 = 6$

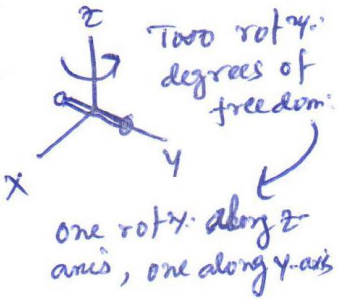


Translational degrees of freedom $= 3$

Energy contribution $= 3 \times \frac{1}{2} kT$

Rotational degrees of freedom $= 2$

Energy contribution $= 2 \times \frac{1}{2} kT = kT$



vibrational motion along the molecular axis.

vibrational degrees of freedom $= 1$

Energy contribution $= kT$

Total energy $= \frac{3}{2} kT + kT + kT = \frac{7}{2} kT$

Total Molar energy $= \frac{7}{2} RT$

$$C_V = \left(\frac{\partial E}{\partial T}\right)_V = \frac{7}{2} R \simeq 7 \text{ cal mol}^{-1} K^{-1}$$

$C_P = 9$ cal mol^{-1} K^{-1}

Law of Equipartition of Energy: C_P and C_V Calculation

Calculate the C_P and C_V values of CO_2, SO_2 and NH_3 :-

For CO_2: Total d.o.f $= 3 \times 3 = 9$

Tr. d.o.f $= 3$; Energy $= \frac{3}{2} kT$

Rot. d.o.f $= 2$; Energy $= 2 \times \frac{1}{2} kT = kT$

Vib. d.o.f $= (3 \times 3 - 5) = 4$; Energy $= 4 kT$

Total Energy $= \frac{3}{2} kT + kT + 4 kT = \frac{13}{2} kT$

Molar " $= \frac{13}{2} RT$

$C_V = \left(\frac{\partial E}{\partial T}\right)_V = \frac{13R}{2} \simeq 13$ cal mol^{-1} K^{-1}

$C_P \simeq 15$ cal mol^{-1} K^{-1}

For SO_2:- Total d.o.f $= 3 \times 3 = 9$

Tr. " $= 3$; Energy $= \frac{3}{2} kT$

Rot. " $= 3$; Energy $= \frac{3}{2} kT$

Vib. " $= (3 \times 3 - 6) = 3$; Energy $= 3 kT$

Total energy $= \frac{3}{2} kT + \frac{3}{2} kT + 3 kT = 6 kT$

Molar " $= 6RT$

$C_V \simeq 6R \simeq 12$ cal mol^{-1} K^{-1}

$C_P \simeq 14$ cal mol^{-1} K^{-1}

For NH_3:- Total d.o.f $= 3 \times 4 = 12$

Tr. " $= 3$; Energy $= \frac{3}{2} kT$

Rot. " $= 3$; " $= \frac{3}{2} kT$

Vib. " $= (3 \times 4 - 6) = 6$; Energy $= 6 kT$

Total energy $= \frac{3}{2} kT + \frac{3}{2} kT + 6 kT = 9 kT$

Molar energy $= 9 RT$

$C_V = 9R \simeq 18$ cal mol^{-1} K^{-1} ; $C_P \simeq 20$ cal mol^{-1} K^{-1}

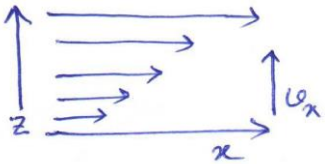
Limitations of Equipartition Principle:- **Excluding monoatomic gases C_P and C_v values calculated on the basis of equipartition principle matches with the experimental values only at high temp.**

Why?

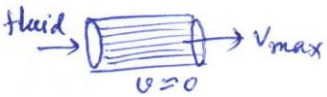
- ✓ **Equipartition principle is derived from Maxwell distribution of molecular speed which is a classical theory.**
- ✓ **This principle is not applicable for systems having quantized energy level or in other word having quantized motion like rot. or vib. (i.e., diatomic or polyatomic gases).**
- ✓ **At high temp, the quantized energy level becomes continuous. As a result, calculated C_p, C_v values based on classical theory matches with experimental values.**

Viscosity of Gases

Viscosity is a flowing property associated with fluids. When a fluid moves under a low pressure difference, it flows in layer (streamline motion). Top most layer has maximum velocity while bottom most layer adjacent to the solid surface remains almost stationary (no-slip condition). When the flow attains steady state condition, there occurs definite velocity gradient, $\frac{dv}{dz}$.



The fluid layers move along X-axis in XY-plane and velocity of the layers are increasing along Z-axis.



Similarly, when a fluid flows in a tube, center layer has maximum velocity and the later attached to the walls of the tube has minimum velocity.

Adjacent horizontal layers of fluid flow at different velocities and slide over one another. As the two adjacent layers slip past each other, each exerts a frictional resistive force on the other, and this internal friction gives rise to viscosity.

Thus, **viscosity is the internal friction developed between two fluid layers moving with unequal velocities.** Due to this property, some fluids flow more easily than the others. This property that characterizes a fluid's resistance to flow is called its viscosity.

f = viscous force

$f \propto A$ (Area of contact)

$\propto dv$ (velocity difference of two adjacent layers)

$\propto \frac{1}{dz}$ (dz = distance betwn. two layers)

Thus, $f \propto A \cdot \frac{dv}{dz}$ $\left(\frac{dv}{dz} = \text{velocity gradient}\right)$

$f = -\eta A \frac{dv}{dz}$; − sign indicates that viscous force acts in opposition to the fluid flow.

η = viscosity coefficient of the fluid, depends on the nature of the fluid and temp.

Viscosity of Gases

Definition of Viscosity Co-efficient :-

$$|f| = \eta A \frac{dv}{dz}$$

$$\therefore A = 1, \quad dv = 1, \quad dz = 1 \; ; \quad |f| = \eta$$

Viscosity co-efficient is defined as the frictional force required to maintain unit difference in velocities between two fluid layers of unit contact area and which are unit distance apart.

unit of η :- $\eta = |f| \cdot \frac{1}{A} \cdot \frac{dz}{dv}$

CGS system: $= \text{dyne} \cdot \frac{1}{cm^2} \cdot \frac{cm \cdot s}{cm}$

$= \text{dyne} \cdot s / cm^2 = \text{Poise}$ (scientist Poiseuille)

SI system: Newton · s/meter^2 = Pascal sec (Pa s)

$1 \text{ Pa s} = 1 \text{ N m}^{-2} \text{s} = 10^5 \text{dyne} \times (10^2 \text{ cm})^{-2} \text{ sec} = 10 \text{ dyne cm}^{-2} \text{ sec} = 10 \text{ Poise}$

Dimention of η :- $\eta = \frac{f}{A \cdot \frac{dv}{dz}} = \frac{MLT^{-2}}{L^2 \frac{LT^{-1}}{L}} = \frac{M}{LT}$

Relation between viscosity co-efficient and mean free path (l) :-

Viscosity co-eff. (η) can be expressed by following equ.;

$$\eta = \frac{1}{3} \rho \bar{c} l \quad \text{where } l = \text{mean free path.}$$

Replacing $\bar{c} = \sqrt{\frac{8RT}{\pi m}}$; $l = \frac{1}{\sqrt{2} \pi \sigma^2 n}$ and $\rho = n m$,

we will get $\eta = \frac{2}{3} \frac{\sqrt{MRT}}{\pi^{3/2} \sigma^2 N_A}$

Thus, by determining η of a gas at a given temp., it is possible to calculate the molecular diameter, σ , of the gas.

Viscosity of Gases

Effect of temperature and pressure on the viscosity of gases:-

$$\eta = \frac{2}{3}\frac{\sqrt{MRT}}{\pi^{3/2}\sigma^2 N_A}$$

The expression of the viscosity of gases from the kinetic model shows that the viscosity of a gas increases with the increase of temp., $\eta \propto \sqrt{T}$

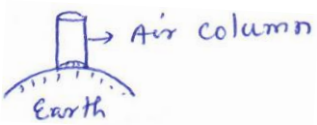
This is qualitatively right and supported by the experiment. This indicates the success of kinetic model of the gas viscosity.

However, in reality, with increase of temp., η of the gas increases more than predicted by the relation. This is due to the assumption of the kinetic theory that the molecules are rigid sphere and have no intermolecular attraction.

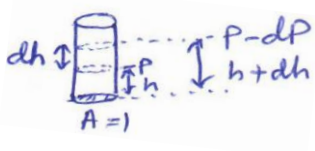
So, Sutherland improved the theory.

Barometric Distribution Law:-

Let us consider an air column of unit cross-section area over the earth surface.



Let the Pr. at height h be P and (h + dh) it is (P – dP)



Change in Pr. $= P - dP - P$

$= -dP$

$= dh\,\rho g$ [ρ = density of the gas inside the slice]

$$-dP = dh\,\rho g = dh \cdot \frac{PM}{RT} \cdot g$$

$$\Rightarrow -\frac{dP}{P} = \frac{Mg}{RT}\,dh$$

Integrating; $-\int_{P_0}^{P} \frac{dP}{P} = \int_0^h \frac{Mg}{RT}\,dh$

For ideal gas,

$$PV = nRT = \frac{w}{M}RT$$ [w = mass of the gas, M = Molar mass]

$$\Rightarrow PM = \frac{w}{V}RT = \rho RT$$

$$\Rightarrow \rho = \frac{PM}{RT}$$

$$\Rightarrow \ln\frac{P}{P_0} = -\frac{Mgh}{RT}$$

P_0 = Pr. at earth surface

P = " " altitude h

$$\therefore P = P_0\, e^{-Mgh/RT}$$

→ Barometric distribution law

Boltzmann Distribution Law

$P = P_0 e^{-Mgh/RT}$ Again, $P \propto n$ [n = no. of molecules per unit volume

$P_0 \propto n_0$ n_0 = " " " " " " at earth surface]

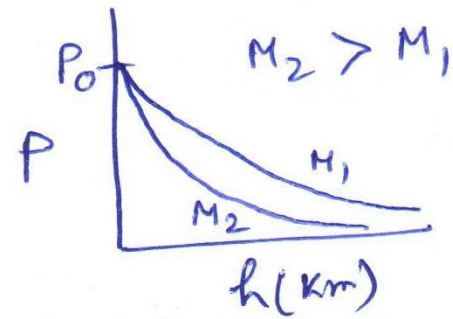
$\frac{P}{P_0} = \frac{n}{n_0}$; Thus, $n = n_0 e^{-Mgh/RT}$ → Boltzmann distribution law.

Features of Barometric Distribution Law:-

(1) Pressure decreases exponentially with altitude

(2) The exponential decrease will be more steeper for gas with higher molecular mass

(3) At a particular altitude the pressure would be more at higher temp



Maxwell-Boltzmann Distribution Law:-

Maxwell distribution law of molecular speed: $dn_c = 4\pi n \left(\frac{m}{2\pi kT}\right)^{3/2} c^2 e^{-mc^2/2kT} dc$

Boltzmann distribution law: $n = n_0 e^{-Mgh/RT} = n_0 e^{-mgh/kT}$

Replacing n in Maxwell distribution;

$$dn_c = 4\pi n_0 e^{-mgh/kT} \left(\frac{m}{2\pi kT}\right)^{3/2} \cdot c^2 \cdot e^{-mc^2/2kT} \cdot dc$$

$$= 4\pi n_0 \left(\frac{m}{2\pi kT}\right)^{3/2} c^2 e^{-(\frac{1}{2}mc^2 + mgh)/kT} \cdot dc$$

$$= 4\pi n_0 \left(\frac{m}{2\pi kT}\right)^{3/2} c^2 e^{-E/kT} dc \quad [E = KE + PE]$$

$e^{-E/kT}$ = Boltzmann factor